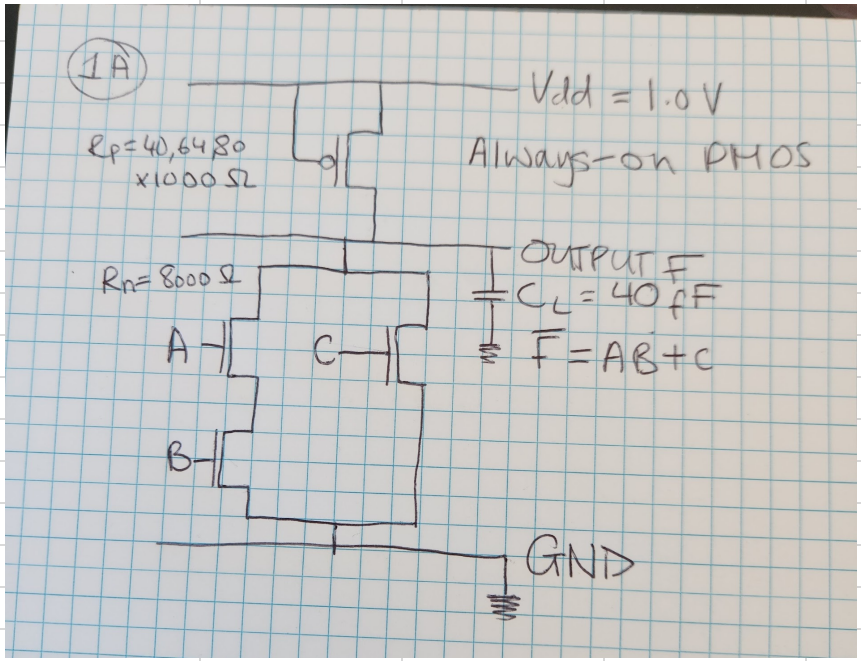
1A



I apologise for the trouble! Due to my disability, I want to preserve my wrist and hand function so I am typing the work I can. I hope that I have formatted my work in a clear enough manner to mark, but if not, please let me know and I will explain my reasonings over a Zoom call or in person.

VOL ≤0.20 V

Vol = Vdd*Rn/Rtotal

PMOS Rp (ohms)	Vdd (V)	CL (fF)	ABC	Notes	Ohms	Vol
40000	1	40	110	AB sequence, 2Rn.	16000	0.2857142857
40000	1	40	`011`	C sequence, Rn	8000	0.1666666667
40000	1	40	111	AB and C,	5333.333333	0.1176470588

PMOS Rp (ohms)	Vdd (V)	CL (fF)	ABC	Notes	Ohms	Vol
64000	1	40	110	AB sequence, 2Rn.	16000	0.2
64000	1	40	`011`	C sequence, Rn	8000	0.1111111111

	64000	1	40	111	AB and C,	5333.333333	0.07692307692
	PMOS Rp (ohms)	**Vdd (V)**	**CL (fF)**	**ABC**	**Notes**	**Ohms**	**Vol**
	80000	1	40	110	AB sequence, 2Rn.	16000	0.1666666667
	80000	1	40	`011`	C sequence, Rn	8000	0.09090909091
	80000	1	40	111	AB and C,	5333.333333	0.0625
	The pull up resistance of 80000 perfectly satisfies the Vol requirement of 0-outputs having a Vol output of less than 0.2. As the requirement is that Vol is less than or equal to 0.2, 64000 ohms also squeaks by.						
	Pstatic = Vdd*I		V=IR	I=V/R	P = 1*1/R		
	PMOS Rp (ohms)	**Ohms**	**Vol**	Vdd	P (watts)		
	64000	5333.333333	0.07692307692	1	0.000014423076		14.4 Microwatts
	PMOS Rp (ohms)	**Ohms**	**Vol**	Vdd	P		
	80000	5333.333333	0.0625	1	0.00001171875		11.7 Microwatts
	Vol is affected by the ratio between resistors - think of the VTC curve! - and static power is over the whole circuit and depends on all the resistors summed.						
1B							
	A=1	C=0	B = 0 --> 1 --> 0		Circuit transition from on to off to on		
	Pull up resistances: 64000 and 80000 Ohms				Vdd --> Vol --> Vdd because PMOS always on		
					100-->110-->100		
	FALLING PROPAGATION: ON TO OFF						
	Resistance pulldown = 16000 ohms, 8000 + 8000				Vout(t) = Vpost+ (Vpre-Vpost)*e^(-time/tau)		
	Rp active, Rn active.	Equivalent resistance:	(RP*RN)/ (RP+RN)				

PMOS Rp (ohms)	**Ohms**	R equiv	Vol	Vdd	CL	Tau
64000	5333.333333	4923.076923	0.2	1	40*10^-15	0.0000000001969230769
		Ohms				
0.5 target V	equals	0.2 + (1-0.2)*e^(-time/tau)				
0.3/0.8	equals	e^(-time/tau)				
-0.980829253	equals	-time/tau				
0.0000000001931479144		=time	equals	193 picoseconds		
PMOS Rp (ohms)	**Ohms**	R equiv	Vol	Vdd	CL	Tau
80000	5333.333333	5000	0.1666666667	1	40*10^-15	0.0000000002
		Ohms	V			
0.5 target V	equals	1/6+ (1-1/6)*e^(-time/tau)				
2/5	equals	e^(-time/tau)				
-0.9162907319	equals	-time/tau				
0.0000000001832581464		=time	equals	183 picoseconds		(unsure)
RISING PROPAGATION: OFF TO ON						
Current depends only on always-on pmos and the nmoses turning off						
PMOS Rp (ohms)	**Ohms**	R equiv	Vol	Vdd	CL	Tau = Rp CL
64000	5333.333333	4923.076923	0.2	1	40*10^-15	0.00000000256
Vout	equals	Vfinal + (Vpre - Vpost) * e^(-time/tau)				
0.5	equals	1+(0.2-1)*e^(-time/tau)				
-0.5/(-0.8)	equals	e^(-time/tau)				
0.4700036292	equals	time/tau				

0.000000001203209291			equals	1203 picoseconds		
PMOS Rp (ohms)	**Ohms**	R equiv	Vol	Vdd	CL	Tau = Rp CL
80000	5333.333333	5000	0.1666666667	1	40*10^-15	0.0000000032
Vout	equals	Vfinal + (Vpre - Vpost) * e^(-time/tau)				
0.5	equals	1+(1/6-1)*e^(-time/tau)				
-0.5/(5/6)	equals	e^(-time/tau)				
0.5108256238	equals	time/tau				
0.000000001634641996		equals	1634 picoseconds			
Delay:						
For 64k ohms, falling delay is 193 picoseconds and the rising delay is 1203 picoseconds. For 80k ohms PMOS, falling delay is 183 picoseconds and rising delay is 1634 picoseconds. For a timing-critical application, I would go for the smaller PMOS resistor to minimise the rising time as the difference between PMOS resistors there is far greater than the difference in falling times.						
Static Power:						
For the worst case, where the path of current is two resistors in series, the 64k ohm PMOS loses 14.4 microwatts and the 80k ohm PMOS loses 11.7 Microwatts. For a timing constrained circuit, I would choose the bigger PMOS transistor as the amount of power lost is less.						